Applications of integral test

1. Harmonic series.

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

By the integral test $\sum_{n=1}^{\infty} 1/n$ converges if and only if $\int_1^{\infty} \frac{1}{t}\,dt$ is finite.

$$\int_1^{\infty} \frac{1}{t}\,dt = \lim_{T\to\infty} \int_1^{T} \frac{1}{t}\,dt$$

$$= \lim_{T\to\infty} \left[\ln(t)\right]_1^{T}, \text{ standard integral}$$

$$= \lim_{T\to\infty} \ln(T), \text{ since } \ln(1) = 0$$

thus is a divergent limit, $\ln(T)$ is unbounded as $T\to\infty$

So $\int_1^{\infty} \frac{1}{t}\,dt = \infty$

and so $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent also.

Can extend this result to the family of hyper-harmonic series.

$$\sum_{n=1}^{\infty} \frac{1}{n^a}, \quad \text{for some } a \in \mathbb{R}^+$$

Some particular cases

$a = 0$, $\sum \frac{1}{n^0} = \sum \frac{1}{1} = \sum_{n=1}^{\infty} 1$, divergent.

$a = 1$, harmonic series we've just discussed

otherwise. $(a \neq 0, 1)$

$$\int_1^{\infty} \frac{1}{t^a}\, dt = \int_1^{\infty} t^{-a}\, dt.$$

$$= \lim_{T \to \infty} \left[\frac{1}{-a+1} t^{-a+1} \right]_1^T$$

$$= \lim_{T \to \infty} \left(\frac{T^{-a+1}}{-a+1} - \frac{1}{-a+1} \right)$$

$\longrightarrow 0$ as $T \to \infty$

as long as $-a+1 < 0$

and is divergent if $-a+1 > 0$

So $\int_1^{\infty} \frac{1}{t^a}\, dt$ $\begin{cases} = \infty \text{ for } a < 1 \\ \text{finite for } a > 1 \end{cases}$

By the integral test, the hyperharmonic series behaves in a similar.

$$\sum_{m=1}^{\infty} \frac{1}{m^a}$$

$$\begin{cases} \text{convergent} & \text{for } a > 1 \\ \text{divergent} & \text{for } 0 < a \leq 1 \end{cases}$$

So $\sum_{m=1}^{\infty} \frac{1}{m}$ is the border between convergent and divergent behaviour of these hyper-harmonic series.

Comparison Tests

Judging convergence / divergence by comparing to a known series (hyper-harmonic, geometric series, ...)

Standard Comparison Test

Pf:

If $a_m \leq c_m$.

$$\Rightarrow \sum_{m=1}^{k} a_m \leq \sum_{m=1}^{k} c_m.$$

and the $\left\{ \sum_{m=1}^{k} c_m \right\}_{k=1}^{\infty}$ is an increasing sequence, converging to c, (can hence bounded above by c).

$$\Rightarrow \left\{ \sum_{m=1}^{k} a_m \right\}_{k=1}^{\infty} \text{ is also}$$

increasing and bounded above by c.

$\Rightarrow$ converges, by the Monotone Conv. Th.

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent.

Note: If we establish:

$a_n \geq c_n$, for all $n \geq 1$

OR $a_n \leq d_n$, for all $n \geq 1$.

then we cannot draw a conclusion about $\sum_{n=1}^{\infty} a_n$

An example

Consider $\sum_{n=1}^{\infty} \frac{1}{3^n + n^2}$

certainly $\frac{1}{3^n + n^2} \to 0$ as $n \to \infty$

Notice that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent hyper-harmonic series, and $\sum_{n=1}^{\infty} \frac{1}{3^n}$ is a convergent geometric series; and so we could take either of these as comparison series, since:

for all $n \geq 1$ $\quad \frac{1}{3^n + n^2} < \frac{1}{n^2}$

and so by the comparison test $\sum_{n=1}^{\infty} \frac{1}{3^n + n^2}$ converges also.

Often the limit comparison test can be easier to apply.

Consider $\sum_{n=1}^{\infty} a_n$ where $a_n = \dfrac{\sqrt{n}+3}{n^2-n+1}$

How to choose the comparison series?

Consider the dominant behaviour "in n" in this term, for very large n.

$$a_n \approx \frac{\sqrt{n}}{n^2} = \frac{n^{1/2}}{n^2} = \frac{1}{n^{3/2}}$$

So choose $\frac{1}{n^{3/2}}$ as our comparison term, and note that $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$, a convergent hyper-harmonic series by since $\frac{3}{2} > 1$.

Let's execute the test.

$$\lim_{n\to\infty} \frac{a_n}{1/n^{3/2}} = \lim_{n\to\infty} \left(\frac{\sqrt{n}+3}{n^2-n+1} \cdot n^{3/2} \right)$$

$$= \lim_{n\to\infty} \left(\frac{n^2 + 3n^{3/2}}{n^2-n+1} \right)$$

$$= \lim_{n\to\infty} \left(\frac{1 + 3n^{-1/2}}{1 - \frac{1}{n} + \frac{1}{n^2}} \right)$$

$$= \frac{\lim_{n\to\infty}(1 + 3n^{-1/2})}{\lim(1 - 1/n + 1/n^2)}$$, by the algebra of limits theorem

$= \frac{1}{1} = 1$, since $\frac{1}{n}, \frac{1}{n^2}, \frac{1}{n^{1/2}} \to 0$ as $n \to \infty$

$L = 1$

and by the algebra of limits theorem.

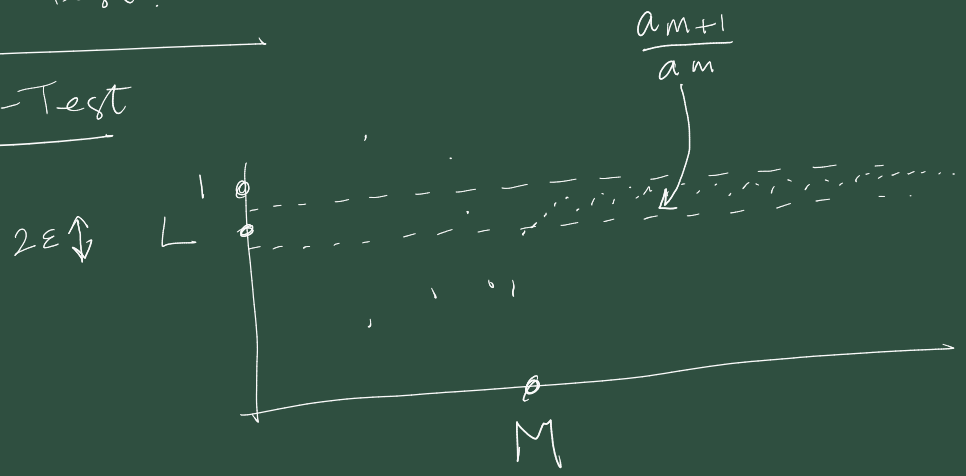
So by the limit comparison test.

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}+3}{n^2-n+1}$$ is a convergent series also.

Continue at 12:00

Leave you to read up on Root Test, Alternating series Test.

Ratio-Test



$$\Rightarrow \text{for } m \geq M \quad \frac{a_{m+1}}{a_m} < \frac{(L+1)}{2} < 1.$$

$$\Rightarrow \text{for } m \geq M \quad a_{m+1} < \frac{(L+1)}{2} a_m$$

$$\sum_{m=1}^{\infty} a_m = \underbrace{(a_1 + a_2 + \ldots + a_M)}_{\text{some finite number}} + \sum_{n=1}^{\infty} a_{M+n}$$

$\Rightarrow \sum_{m=1}^{\infty} a_m$ converges

---

if $L = 1$



$\frac{a_{m+1}}{a_m} < 1$? > 1 ?

we don't know from just the fact that

$\frac{a_{m+1}}{a_m} \to 1$ as $m \to \infty$

$a_{m+1} > a_m$?

$< a_m$?

So $L = 1$ case remains inconclusive.

Applications of ratio test.

Can typically be used where the ratio of neighbouring terms $\frac{a_{m+1}}{a_m}$ undergoes significant simplification (eg. depends on exponentials, factorials, ...)

1. Consider $\sum_{n=1}^{\infty} \frac{2}{n!}$, $a_n = \frac{2}{n!}$.

Hopefully we feel this is convergent, since $\frac{2}{n!} \to 0$ as $n \to \infty$ and it does so quite fast as factorials increase very fast.

Consider $\lim_{n \to \infty} \left(\frac{2}{(n+1)!} \cdot \frac{n!}{2} \right)$ ← $\frac{a_{n+1}}{a_n}$

$= \lim_{n \to \infty} \left(\frac{1}{n+1} \right)$

$L = 0$

So by the ratio-test $\sum_{n=1}^{\infty} \frac{2}{n!}$ is convergent.

---

2. Consider: $\sum_{n=1}^{\infty} a_n$, where $a_n = \frac{3^n}{n^3}$

actually a_n looks unbounded as $n \to \infty$.

so surely this is divergent? .... Can be shown formally using ratio-test.

$$\lim_{n \to \infty} \left(\frac{a_{n+1}}{a_n} \right) = \lim_{n \to \infty} \left(\frac{3^{n+1}}{(n+1)^3} \cdot \frac{n^3}{3^n} \right)$$

$$= \lim \left(3 \cdot \frac{n^3}{(n+1)^3} \right)$$

$$= 3 \lim_{n\to\infty} \left(\left(\frac{n}{n+1} \right)^3 \right)$$

$$= 3 \lim_{n\to\infty} \left(\left(\frac{1}{1+1/n} \right)^3 \right)$$

$L = 3$.

So by the ratio test $\sum_{n=1}^{\infty} \frac{3^n}{n^3}$ is divergent ($L = 3 > 1$).

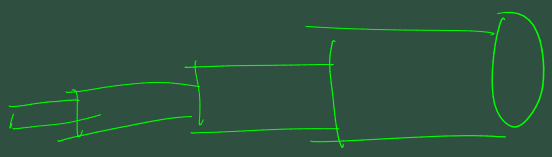
Let's try to analyze the harmonic series using the ratio-test.

$$\sum_{n=1}^{\infty} \frac{1}{n}.$$

$$\lim_{n\to\infty} \left(\frac{1}{n+1} \cdot n \right) = \lim_{n\to\infty} \frac{n}{n+1}$$

$$= 1.$$

So the ratio-test can not reach a conclusion on the harmonic series.



Telescoping series A class of series where a lot of cancellation happens in the partial sums, so that we can determine the actual sum of the series. ✓ coming in section 3.3

Uses the 'partial fraction expansion' technique.

or rational functions (= ratio of polynomials).

Show this by example Q14 from the tutorial sheet.

Consider $\sum_{n=1}^{\infty} b_n$, where $b_n = \frac{8}{n^2+6n+5}$

$$b_n = \frac{8}{n^2+6n+5} = \frac{8}{(n+5)(n+1)}$$

it turns out we can rewrite b_n in the form

$$b_n = \frac{\alpha}{n+5} + \frac{\beta}{n+1},$$ so called partial fraction expansion.

for some choice of $\alpha, \beta \in \mathbb{R}$.

$$= \frac{\alpha(n+1) + \beta(n+5)}{(n+5)(n+1)}$$

$$= \frac{(\alpha+\beta)n + (\alpha+5\beta)}{(n+5)(n+1)}$$

Compare this numerator to the original numerator 8.

$$(\alpha+\beta)n + (\alpha+5\beta) = 8$$

for all n?

$$\Rightarrow \left\{ \begin{array}{l} \alpha+\beta = 0 \\ \alpha+5\beta = 8 \end{array} \right\}$$

In general we might need Gaussian Elimination.

$$\Rightarrow \quad \begin{cases} \beta = -\alpha \\ \alpha - 5\alpha = 8 \end{cases}$$

$$\Rightarrow \quad -4\alpha = 8 \Rightarrow \alpha = -2 \text{ & } \beta = 2$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{-2}{n+5} + \frac{2}{n+1} \right)$$

$$= \lim_{k \to \infty} \left(\sum_{n=1}^{k} \left(\frac{-2}{n+5} + \frac{2}{n+1} \right) \right)$$

$$= \lim_{k \to \infty} \Bigg[\left(\frac{-2}{6} + \frac{2}{3} \right) + \left(\frac{-2}{7} + \frac{2}{3} \right) + \left(\frac{-2}{8} + \frac{2}{4} \right) + \left(\frac{-2}{9} + \frac{2}{5} \right) + \left(\frac{-2}{10} + \frac{2}{6} \right) + \left(\frac{-2}{11} + \frac{2}{7} \right) + \ldots + \ldots + \left(\frac{-2}{k+4} + \frac{2}{k} \right) + \left(\frac{-2}{k+5} + \frac{2}{k+1} \right) \Bigg]$$

Notice that each negative term cancels with a similar positive term coming four steps later. So almost all these fractions cancel.

$$\lim_{k \to \infty} \left(\frac{2}{2} + \frac{2}{3} + \frac{2}{4} + \frac{2}{5} - \frac{2}{k+5} - \frac{2}{k+4} - \frac{2}{k+3} - \frac{2}{k+2} \right)$$

$= \frac{2}{2} + \frac{2}{3} + \frac{2}{4} + \frac{2}{5}$, since $\frac{1}{k+x} \to 0$ as $k \to \infty$, for $x = 5, 4, 3, 2$.

$= \frac{60 + 40 + 30 + 24}{60}$

$= \frac{154}{60} = \frac{77}{30}$

and algebra of limits theorem

So $\sum_{n=1}^{\infty} b_n = \frac{77}{30}$

Leave you to read 2.4,